Introduction to Recommender Systems

Ariton Verush, Exercise 6

Task 4: Long tail plot - Lines 38 to 43: This plot is used to explore popularity patterns in useritem interaction data such as clicks, ratings, or purchases. Typically, only a small percentage of items have a high volume of interactions, and this is referred to as the “head”. Most items are in the “long tail”, but they only make up a small percentage of interactions. Observe the graph generated when running the code and answer the following questions:

How will you describe the 1482 items that are in the head of the graph?

• Head items (1482): These are the most frequently interacted with and highly watched movies, like critically acclaimed films and so on. The head represents the top rank of popular movies.

How will you describe the 20534 items that are in the tail of the graph?

• Tail items (20534): These include the less well-known and not so popular/interacted with movies that haven't achieved widespread attention. The long tail offers a range of unique and undiscovered cinematic options.

• If a RS recommends items that are only in the head of the graph, what can you say about this RS in terms of personalization and serendipity?

RS recommending only head items: Such a recommender system would lack true personalization, as it merely suggests the most popular movies without tailoring to individual user preferences. Furthermore, it would have low serendipity because users are likely already aware of the titles.

Task 5: MEAN AVERAGE RECALL @ K - Lines 126 - 151: A recommender system typically produces an ordered list of recommendations for each user in the test set. MAR@K gives insight into how well the recommender is able to recall all the items the user has rated positively in the test set. Observe and analize the MAR@K graph and answer the following:

• Given CF's better MAR@K: Yes, the higher recall indicates that the CF model is more effective in identifying items relevant to each user compared to random and popular models. This leads to more precise and useful recommendations.

• Better for personalization & serendipity: CF appears to excel in personalization because it has higher relevance of its recommendations. For serendipity, it's less clear—high recall could mean fewer novel suggestions if it merely repeats the user's known preferences. A good CF model can balance both relevance and novelty.

Task 6: Coverage - Lines 134 - 151: Coverage is the percent of items in the training data the model is able to recommend. Observe and analize the Coverage graph and answer the following:

• Random higher coverage: Random recommendations do not prioritize popularity, so they can suggest both well-known and obscure items equally. This ensures a more even distribution and that no item is overlooked.

• Better for personalization & serendipity: CF is likely superior for personalization because it considers user preferences. For serendipity, random has an advantage due to its unpredictability.

Task 7: Novelty - Lines 154 - 160: Novelty measures the capacity of recommender system to propose novel and unexpected items which a user is unlikely to know about already. You will find the results of the novelty measure on the console. Answer the following:

• Better for personalization & serendipity: For personalization, CF is effective through similarity matching. For serendipity, random might give us more unexpected items, but CF can also recommend new, yet relevant items. Balancing familiarity and surprise is essential for a better recommendation experience.

Task 8 Which model will you define as the overall best? The Collaborative Filtering (CF) model can be considered the overall best, as it balances personalization with occasional serendipity.

• When you think exclusively in terms of personalization and serendipity, which one do you consider the best model?

For personalization and serendipity, CF is better because it provides tailored recommendations while also introducing new, relevant items.

• If you have to pick one to implement in a real-life movies recommendation application, which model would you select to implement and why?

For a real-life movie recommendation application, the CF model would be chosen due to its ability to deliver personalized and engaging recommendations, very similar to the preferences of a user.